

Khawaja Yunus Ali University

School of Science and Engineering

B.Sc. in CSE & EEE

Lesson Plan

Course Title : Physics - II

Course Code : CSE 0533 1201 (CSE) & PHY 0533 1201 (EEE)

Term: Spring – 2024	Date: 29.01.2024	Time: 03.30 PM	Room: 327	Activity: Lecture 9
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Topics to be discussed: Thermodynamic potential, Helmholtz free energy, Enthalpy & Gibbs free energy

Target CLO & PLO : CLO2,3,5 & PLO2,4,5,7

Learning Outcomes : At the end of the class the students will be able to:

- ❖ Discuss the thermodynamic potentials
- ❖ Explain the concept of Internal energy, Helmholtz energy & Gibbs free energy
- ❖ Relate these three with the thermodynamic potentials

Methods of Teaching:

Mode/Activities	Time
Feedback of previous class	10min
Interactive discussion on some natural phenomenon	10min
Lecture Presentation on Topics	30 min
Group activity	25 min
Feedback session & Suggest for Assignment	10 min

Assessment Strategy : Quizzes/Class tests, Class Participation & Assignment

Sample Questions (MCQ):

- 1) Which one is **NOT** a thermodynamic potential?
a) Solar energy b) Internal energy c) Enthalpy d) Helmholtz free energy e) Gibbs free energy
- 2) Identify the correct equation for Enthalpy.
a) $E=TS$ b) $E=TS-G$ c) $E=U+PV$ d) $E=PV-TS$ e) $E=TS-U$

Sample Questions (SAQ):

- 1) What are the thermodynamic potentials?
- 2) What is internal energy? Deduce its expression.
- 3) What is Helmholtz free energy?
- 4) What is Enthalpy?
- 5) What is Gibbs free energy?
- 6) What is the significance of positive and negative changes in Gibbs free energy?
- 7) What do you mean by natural and conjugate variable?

Reference Books:

1. Giasuddin ahmed – Physics for Engineers (Part-2)
2. Halliday Resnick - Fundamental of Physics, 2nd edition

Pre-reading book for next Lecture:

- Md. Ziaul Ahsan – Physics for Engineering (Lecture Series), Pages: 179-235
- Dr. Amir Hussain Khan – Physics 2nd Paper (HSC Text Book), Pages: 290-374